ASSIGNMENT 3

AI Safety Improvement and Experimental Validation: Red Teaming, Robustness, Explainability and Human-AI Interaction

Course: AI Security and Risk Analysis

Course Code: AISRA1301

Programme: Master's Degree — AI+X

Institution: Qazaq AI Research University (QAIRU)

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1. Assignment Overview

Assignment 3 is a direct continuation of the Midterm Project.

During the Midterm, students developed a functional AI prototype, selected a scientific research problem and conducted an initial AI safety or security experiment.

In Assignment 3, students must investigate the identified vulnerabilities in greater depth, implement or systematically evaluate an additional safety or security mechanism, and experimentally determine whether the proposed improvement makes the system safer, more robust or more reliable.

Students must continue using the same research topic and AI system developed during the Midterm. A new project or application is not required.

The primary objective is to move from vulnerability identification to evidence-based safety improvement.

Key Difference Between Midterm and Assignment 3

Midterm	Assignment 3
Develop a functional AI prototype	Improve the existing prototype
Conduct initial safety testing	Perform extended testing
Identify vulnerabilities	Evaluate a mitigation mechanism
Establish baseline results	Compare baseline and improved results
Identify research limitations	Analyze residual risks and trade-offs
Propose possible improvements	Implement or experimentally validate an improvement

2. Learning Objectives

Upon successful completion, students should be able to:

1. Conduct systematic AI red teaming or adversarial evaluation.
2. Identify and explain relevant AI system vulnerabilities.
3. Design, implement or adapt an appropriate safety mitigation.
4. Evaluate system robustness, interpretability and reliability.
5. Analyze risks associated with human-AI interaction.
6. Compare baseline and improved configurations using quantitative metrics.
7. Assess security, performance and usability trade-offs.
8. Document reproducible experimental findings.
9. Explain the scientific significance and limitations of the proposed improvement.

3. Assignment Requirements

Part 1. Review of Midterm Results

Begin with the AI system and experimental findings from the Midterm Project.

Provide a concise description of:

- Research problem and selected scientific paper.
- Existing AI system architecture.
- AI model and experimental environment.
- Previously identified safety or security risks.
- Baseline evaluation results.
- Limitations of the initial experiment.

Identify the principal vulnerability or safety limitation that will be investigated in Assignment 3.

Expected outcome: A clearly defined improvement objective based on the Midterm findings.

Part 2. Extended AI Security and Safety Testing

Conduct additional controlled tests to investigate the selected risk more systematically.

A. LLM and RAG Security

- Prompt injection resistance.
- Jailbreak robustness.
- Unsafe or misleading responses.
- Sensitive information disclosure.
- Untrusted document handling.
- Hallucination and factual reliability.

B. Adversarial Machine Learning

- Adversarial examples.
- Input perturbation robustness.
- Evasion resistance.
- Reliability under distribution shifts.
- Model prediction consistency.

C. AI Agent Security

- Unsafe tool invocation.
- Unauthorized actions.
- Excessive permissions.
- Agent workflow manipulation.
- Indirect prompt injection through external resources.

D. AI Reliability and Availability

- Failure detection and recovery.
- Response consistency.
- System availability.
- Robustness under unexpected inputs.
- Error handling and fallback mechanisms.

All testing must be performed on systems and data that students are authorized to use.

Part 3. Additional AI Safety or Security Mechanism

Students must implement, adapt or systematically evaluate at least one additional safety, security or robustness mechanism that was not evaluated during the Midterm.

Possible mechanisms include:

- Input validation and sanitization.
- Prompt injection detection.
- Output verification.
- Restricted tool permissions.
- Retrieval source validation.
- Safety filters or guardrails.
- Adversarial training or preprocessing.
- Confidence-based abstention.
- Human approval for sensitive actions.
- Improved error handling and fallback mechanisms.
- Model monitoring or anomaly detection.

Students must explain:

10. Why the selected mechanism is appropriate.
11. How it works.
12. How it is integrated into or evaluated with the AI system.
13. Which risks it is intended to reduce.
14. Which limitations and residual risks remain.

Important: Installing a tool alone does not satisfy this requirement. Students must demonstrate and evaluate its actual effect on the selected AI system.

Part 4. Research Tools and Frameworks

Students should continue using the tools selected during the Midterm and select additional tools where technically appropriate.

The following tools are recommended.

Category A. AI Security and Red Teaming

Tool	Research Application	Official Link
NVIDIA garak	LLM vulnerability testing	https://github.com/NVIDIA/garak
Microsoft PyRIT	AI red teaming and attack evaluation	https://github.com/Azure/PyRIT
Promptfoo	LLM security and regression testing	https://www.promptfoo.dev/
Giskard	AI testing and vulnerability evaluation	https://github.com/Giskard-AI/giskard
AgentDojo	AI agent prompt injection evaluation	https://github.com/ethz-spylab/agentdojo
IBM ART	Adversarial machine learning testing	https://github.com/Trusted-AI/adversarial-robustness-toolbox

Category B. AI Defense and Robustness

Tool	Research Application	Official Link
NVIDIA NeMo Guardrails	Safety rules and guardrails for LLM applications	https://github.com/NVIDIA-NeMo/Guardrails
Meta Prompt Guard	Prompt injection and jailbreak detection	https://github.com/meta-llama/PurpleLlama
Meta Llama Guard	Input and output safety classification	https://github.com/meta-llama/PurpleLlama

Meta Code Shield	Detection of insecure generated code	https://github.com/meta-llama/PurpleLlama
IBM ART	Adversarial defenses and robustness evaluation	https://github.com/Trusted-AI/adversarial-robustness-toolbox
Microsoft Presidio	Detection and protection of sensitive information	https://github.com/microsoft/presidio

The Meta tools are part of the Purple Llama ecosystem. Students should consult the individual component documentation and verify model access, licensing and technical requirements before implementation.

Category C. Explainability, Reliability and Human-AI Interaction

Tool	Research Application	Official Link
SHAP	Model explanation and feature attribution	https://shap.readthedocs.io/
LIME	Local explanations of model predictions	https://github.com/marcotcr/lime
Captum	Model interpretability	https://captum.ai/
Evidently AI	Model quality, drift and monitoring	https://www.evidentlyai.com/
Deepchecks	ML validation and monitoring	https://docs.deepchecks.com/
Microsoft HAX Toolkit	Human-AI interaction design	https://www.microsoft.com/en-us/haxtoolkit/
Ragas	RAG application evaluation	https://docs.ragas.io/

Tool Selection Requirement

Students must:

15. Continue using their existing Midterm evaluation environment.
16. Select at least one additional relevant safety, security or robustness mechanism.
17. Use appropriate testing or evaluation tools to measure its effectiveness.
18. Explain why the selected tools are suitable for their research problem.

Students are not required to use every listed tool or install unrelated frameworks.

4. Experimental Validation

Part 5. Baseline and Improved Configuration

Students must compare the original Midterm system with the improved or modified system.

The experiment must include:

19. Research question or hypothesis.
20. Baseline configuration.
21. Improved configuration.
22. Comparable test cases.
23. Evaluation metrics.
24. Recorded experimental results.
25. Comparative analysis.
26. Limitations and residual risks.

Use the same test cases and experimental conditions wherever feasible.

If experimental conditions change, explain how the changes affect the comparison.

Part 6. Quantitative Evaluation

Select metrics appropriate to the research problem.

Examples include:

- Attack Success Rate (ASR).
- Attack detection rate.
- Precision, Recall and F1-score.
- False Positive Rate.
- Robust accuracy.
- Hallucination rate.
- Task success rate.
- Response latency.
- System availability.
- Relevant explainability and user-safety measures.

Students must evaluate both safety improvements and possible negative effects on normal system functionality.

Part 7. Experimental Comparison

Illustrative Example — Prompt Injection Defense

Metric	Baseline System	Improved System
Number of test cases	30	30
Successful attacks	12	4
Attack Success Rate	40%	13.3%
Legitimate task success rate	92%	89%
Average response latency	1.1 s	1.4 s

These values are illustrative only. Students must report their own experimental measurements.

Students must explain:

- Whether the selected mitigation reduced the identified risk.
- Whether the observed improvement is supported by the test results.
- Whether legitimate task performance changed.
- Whether response latency or resource consumption increased.
- Which vulnerabilities remain unresolved.
- What additional experiments would strengthen the conclusions.

5. Interpretability and Availability Analysis

Part 8. Explainability and System Reliability

Address the Week 6 topics in relation to the selected AI system.

Students must analyze:

27. Whether the system's behavior can be meaningfully explained or inspected.
28. How failures or unexpected outputs can be identified.
29. What reliability or availability limitations exist.
30. Whether the proposed mitigation changes model behavior.
31. Whether additional transparency or monitoring is necessary.

Students may use SHAP, LIME, Captum, model traces, response analysis or other relevant methods.

A separate explainability experiment is not required when the method is not applicable, but the limitation must be justified.

6. Safe Human-AI Interaction

Part 9. Human-AI Safety Evaluation

Address the Week 7 topic by evaluating how users interact with the AI system.

Consider:

- User overreliance and automation bias.
- Misleading or overconfident outputs.
- Transparency of system limitations.
- Human review or approval requirements.
- User warnings and safety instructions.
- Fallback and escalation procedures.
- Potential misuse of system outputs.

Students must propose at least one practical human-AI interaction improvement or safety recommendation.

This improvement may be implemented in the prototype or documented as a justified design recommendation.

7. Technical Research Report

Length: 2–3 pages, excluding references.

Language: English.

Format: PDF.

Required structure:

32. Title and Research Objective.
33. Midterm Baseline and Identified Risk.
34. Extended Testing Methodology.
35. Additional Safety or Security Mechanism.
36. Experimental Setup and Evaluation Metrics.
37. Results and Comparative Analysis.
38. Interpretability and Availability Analysis.
39. Safe Human-AI Interaction.
40. Limitations and Future Work.
41. References.

The report must include relevant tables or figures supported by experimental evidence.

8. Assignment Deliverables

Submit the following materials through the LMS.

A. Technical Research Report

- A 2–3-page PDF describing the methodology, additional mechanism and experimental findings.

B. Updated AI Prototype

- Updated source code, notebook or repository containing the evaluated improvement.

C. Experimental Evidence

- Test cases, evaluation results, comparison tables, logs and relevant visualizations.

D. Reproducibility Instructions

- Updated README, dependencies, configurations and instructions for repeating the experiment.

E. AI Use Declaration

- Description of substantive AI assistance used during development, testing or reporting.

9. Assessment Rubric

Assessment Criterion	Points
Risk identification and Midterm baseline analysis	15
Additional safety or security mechanism	25
Experimental retesting and comparative evaluation	30
Critical interpretation, explainability and human-AI interaction	20
Documentation and reproducibility	10
Total	100

10. Academic Integrity and Responsible Testing

Students must:

- Use only authorized AI systems and datasets.
- Accurately report all experimental results.
- Cite reused code, models, methods and scientific literature.
- Disclose substantive AI-generated contributions.
- Avoid fabricated evidence or unsupported performance claims.
- Maintain sufficient records to reproduce the experiment.
- Protect personal and confidential information.

Open-source tools and AI-assisted programming are permitted when properly acknowledged and independently verified.